

Inventory Management & Optimization

Data-driven analysis of 4,500 SKUs to cut holding costs, prevent stockouts, and optimize ordering.

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Project Snapshot

Scope

4,500 SKUs; 14 data attributes

Goals

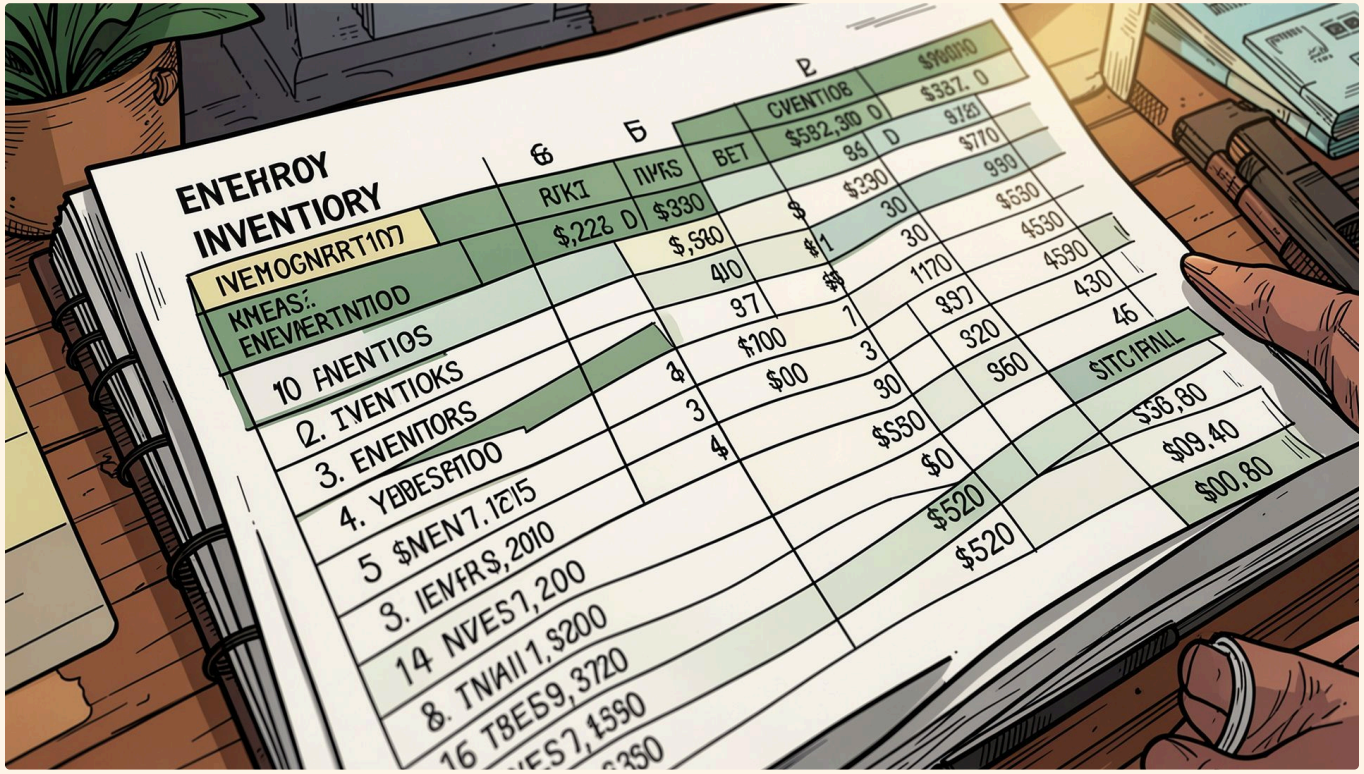
Minimize holding costs · prevent stockouts · optimize EOQ

Tech

Python (Pandas/NumPy), PostgreSQL

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Dataset Highlights



Key Fields

SKU, Category, Unit Cost ($), Annual Demand, Order Cost ($), Holding Cost %, Lead Time (days), Daily Demand, Current Stock, ROP, Safety Stock, Stockouts, ITR

Missing lead times imputed by category median. Standardized prices. Calculated annual consumption value.

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EDA Approach (Python)

01

Load & Clean

Read CSV · impute missing · standardize

02

Feature Engineering

annual_consumption_value = annual_demand × unit_cost

03

ABC Classification

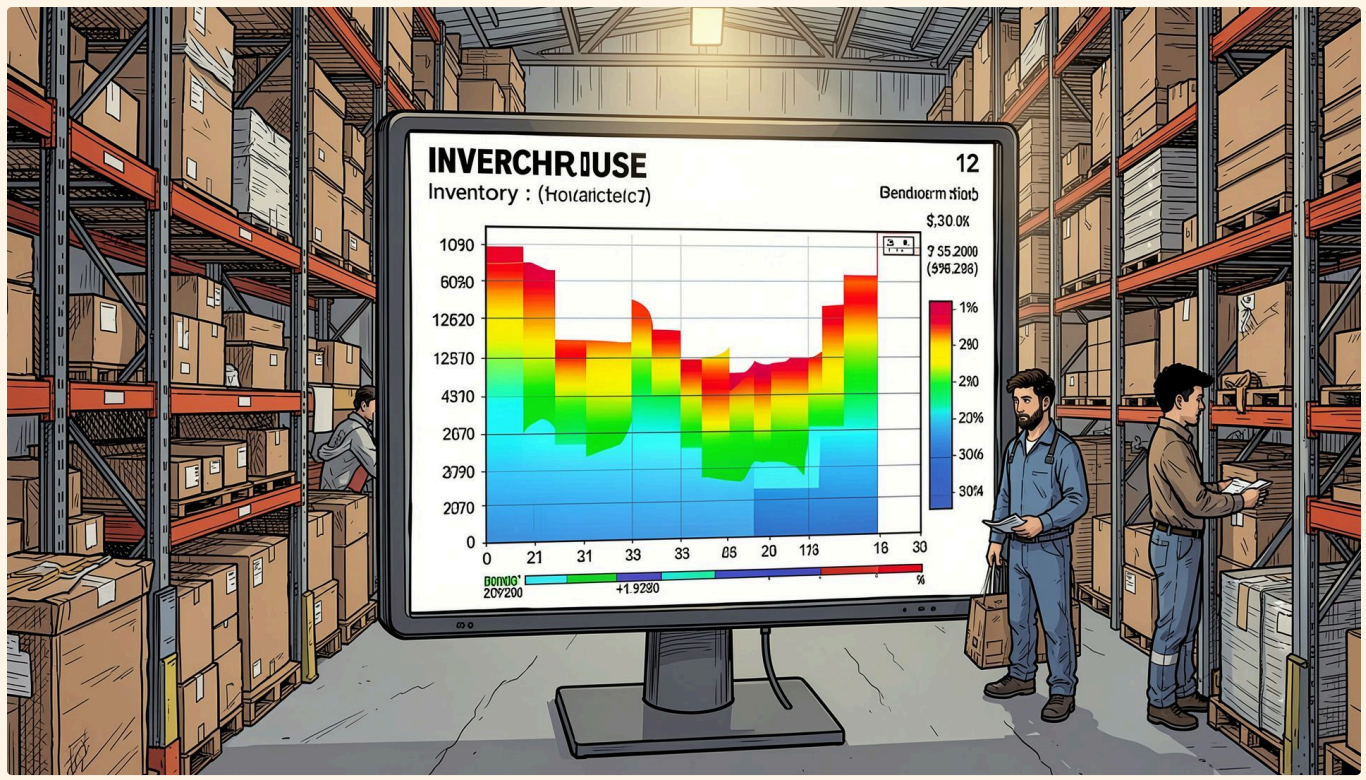
Cumulative % → A ≤70%, B 70–90%, C >90%

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Summary Statistics

Metric	Mean	Std Dev	Median	Max
Unit Cost ($)	48.50	62.10	28.50	450.00
Annual Demand	1,250.40	890.15	1,100.00	5,200.00
Annual Usage ($)	60,644.20	112,450.00	28,500.00	945,000.00
Lead Time (Days)	12.40	4.80	12.00	30.00
Turnover Ratio	5.82	2.15	5.60	14.20

Distribution shows high skew in value and demand. Use ABC to focus controls.



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ABC Results (SQL)

Class A

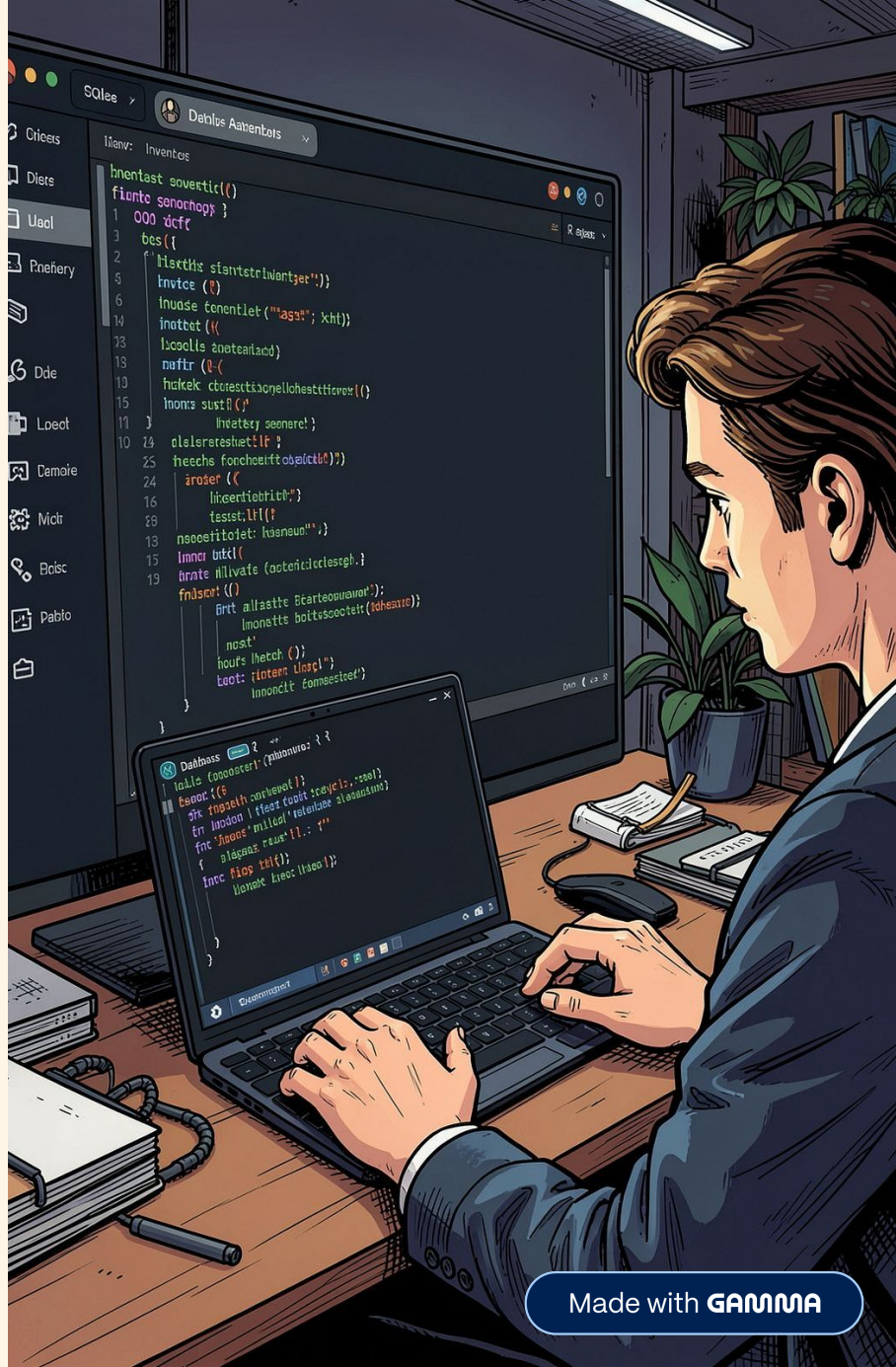
675 SKUs (15%) · $191,029,230 (70%) · Avg ITR 8.45

Class B

1,350 SKUs (30%) · $54,579,780 (20%) · Avg ITR 5.20

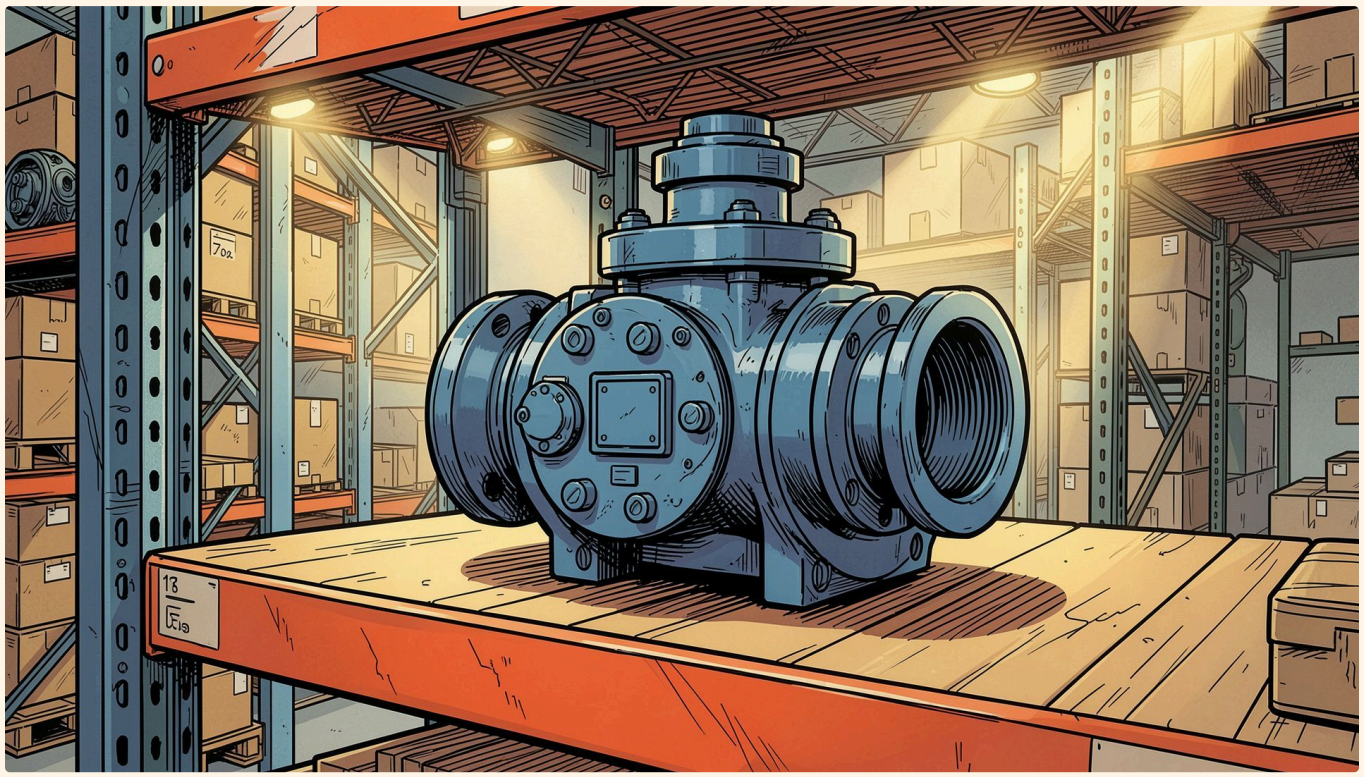
Class C

2,475 SKUs (55%) · $27,289,890 (10%) · Avg ITR 2.85



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High Stockout Risk — Sample Class A SKUs



1

SK-1082

Hydraulic Valve A4 · Current 45 · ROP 180 · Stockouts 12

2

SK-3041

Microcontroller Board V2 · Current 110 · ROP 450 · Stockouts 9

3

SK-0492

Precision Bearing 10mm · Current 85 · ROP 320 · Stockouts 8

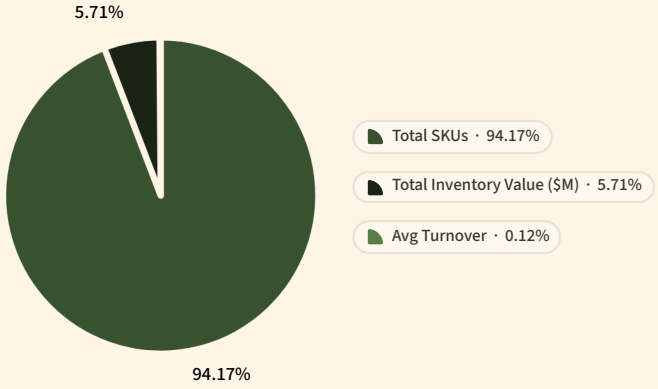
4

SK-2219

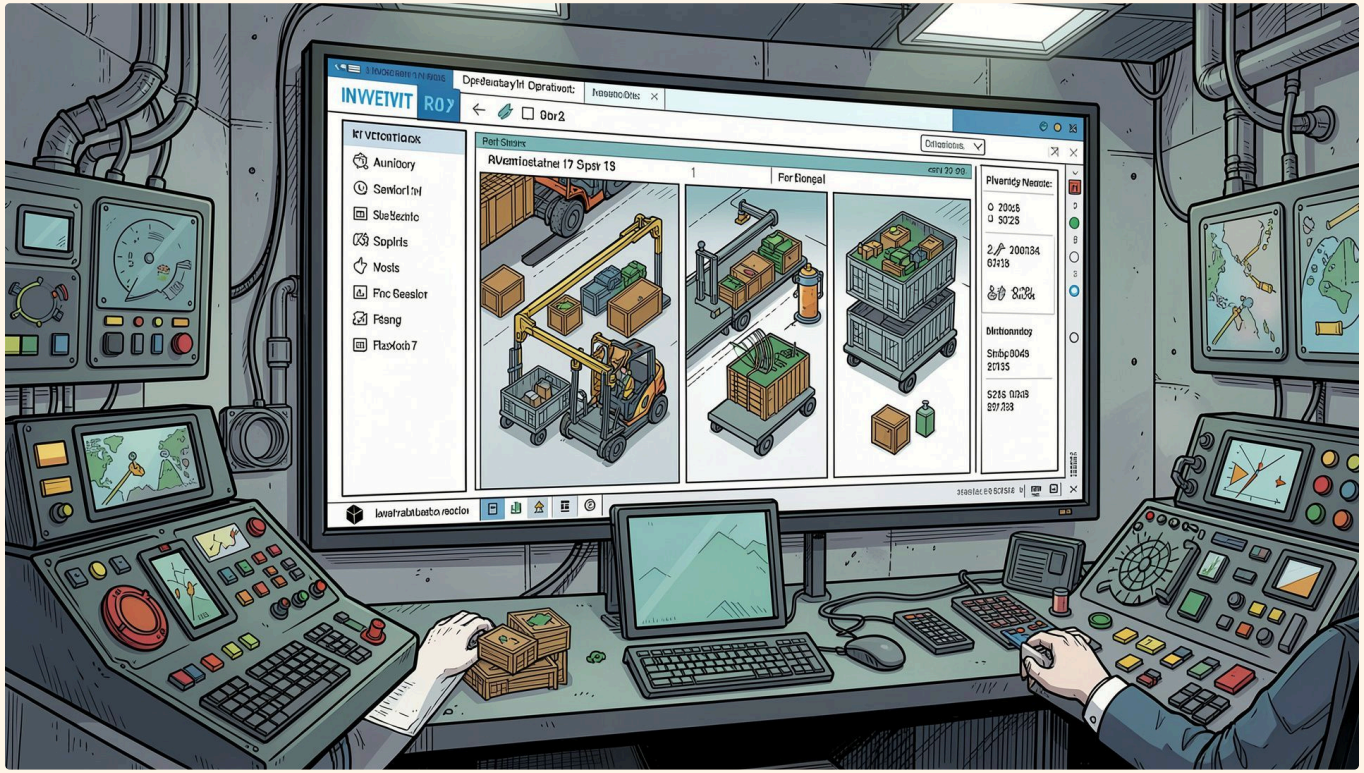
Lithium Cell Pack 12V · Current 60 · ROP 290 · Stockouts 7

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Dashboard Snapshot

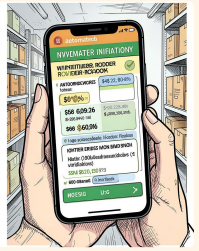


Key KPIs for operations and finance. Visual drill-down recommended by class.



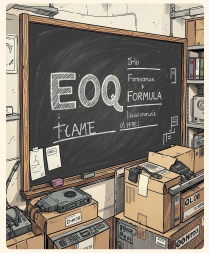
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Optimization Opportunities



Automated ROP

Trigger auto-reorders for Class A when below ROP to avoid stockouts.



EOQ Alignment

Adjust order sizes in Electronics to match EOQ — save up to $1.4M/year in holding costs.



Liquidate Dead Stock

Clear Class C items (turnover <1.5) to free ~$690,000 working capital.

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Next Steps

1 Deploy

Implement automated ROP triggers for Class A

2 Align

Revise purchase orders to EOQ for Electronics

3 Clean

Plan liquidation of slow-moving Class C items



Measure results: holding cost savings, stockout reduction, turnover improvement.

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